

ECON 351 · Focused Practice Set

Set 10 of 10

Chapter 9

Diversification

1 question in this practice set

Questions: Q24

Screenshots from original HW, sample questions, and mock midterm.

Question 1 of 1 (Practice Q24)

Diversification · Sample Q22

Question 21:

Arthur has the following utility from income:

$$u(I) = \sqrt{I}.$$

He has a collection of rare comic books that is worth \$10,000. With probability 50% a silverfish invasion will destroy all his books, bringing the value of his collection to zero. With probability 50% nothing occurs, and his collection keeps its full value. A company called Insect Invasion Insurance (III) offers to partially insure the collection for a price $P = \$1,900$. That is, Arthur first pays the insurance premium $P = \$1,900$; and then, in case of a silverfish invasion, III will send to Arthur a \$2,044 check. If Arthur buys this partial insurance plan, then his expected utility will be _____. Therefore, Arthur _____.

[Be careful with your computation. Keep as many decimal places as possible along all the steps of the problem. Once you are done with all computations, round your final answer to the nearest integer. It should be clear which one.]

- (A) 53 ; should buy the insurance
- (B) 51 ; should buy the insurance
- (C) 51 ; should not buy the insurance
- (D) 53 ; should not buy the insurance
- (E) None of the above, because Arthur is indifferent between buying the insurance or not.

Question 22:

Feng has all her wealth invested in Company A and in Company B. These are agricultural companies, and their value can go up or down depending on the amount of rainfall this season. There is a 50% chance that it

will rain a lot, in which case the value of her investment in Company A will go up to \$100 while her investment in Company B will go up to \$90. There is a 50% chance of a drought, in which case the value of her investment in Company A will go down to \$80, while her investment in Company B will go down to \$70.

Nan has her wealth invested in Company C and in Company D. Company C is an airline company, while Company D is an oil company. Their value can go up or down depending on the international price of oil. There is a 50% chance that the price of oil will go up, in which case her investment in Company D will go up to \$100 while her investment in Company C will go down to \$70. There is a 50% chance that the price of oil will go down, in which case the value of her investment in Company D will go down to \$80, while her investment in Company C will go up to \$90.

- a) Are the values of Company A and Company B positively or negatively correlated?
- b) Are the values of Company C and Company D positively or negatively correlated?
- c) Who has the riskier investment portfolio, Feng or Nan?

Question 23:

Mark wants to buy a new car to become a rideshare driver. He can buy a regular car or an electric car. His profit depends on the type of car he purchases and on whether the future price of gas will be high or low. The following table summarizes his profit for each type of car and for the different gas prices.

	Low Prices	High Prices
Regular Car	\$110	\$80
Electric Car	\$120	\$90

There is a 50/50 chance that the gas price will be low or high. Mark wants to maximize his expected profit.

- a) Should Mark purchase a Regular Car or an Electric Car?

Question 24:

Mark wants to buy a new car to become a rideshare driver. He can buy a regular car or an electric car. His profit depends on the type of car he purchases and on whether the future price of gas will be high or low. The following table summarizes his profit for each type of car and for the different gas prices.

	Low Prices	High Prices
Regular Car	\$110	\$80
Electric Car	\$102	\$90

There is a 50/50 chance that the gas price will be low or high. Mark wants to maximize his expected profit.

- a) Should Mark purchase a Regular Car or an Electric Car?

b) _____
